Empirical Characterization of Health Documentation Governance in Open Source Projects

anonymous

Abstract—**Open Source Software (OSS) health documentation comprises of files that govern how users or contributors join, interact with, and sustain an open source project; these artifacts are essential to project continuous sustainability, yet their governance is not clearly understood. We present a decade-long empirical characterization of documentation governance across 100 open source repositories, examining 37,447 health documentation commits through three lenses: rhythm, intention, and ownership. Our analysis identify three distinct documentation rhythm archetypes (Consistent (37%), Occasional (50%), and Sparse (13%) Documenters), we show that documentation rhythm structure is durable: Consistent and Sparse Documenters persist as the same archetype from one year to the next, while Occasional function mostly as a transitional state repositories pass through rather than settle into. We found ownership to be concentrated in a minority of contributors, tracking these dimensions across the full decade studied, however, shows documentation governance is not static: the breadth of contributor participation and the deliberateness of documentation activity have each declined significantly as projects mature, even as rhythm archetypes remain stable, and self-reinforcing, once established. Critically, we show that this ownership structure is not merely descriptive but also predictive: documentation bus factor independently predicts documentation staleness beyond project size and documentation volume, the breadth of documentation participation independently predicts newcomer retention, and also participation breadth further predicts the trajectory of a project's overall activity, extending beyond documentation itself. We discuss further implications for OSS communities and their sustainability.**

I. Introduction

Open source software (OSS) has become a critical component of global software supply-chain infrastructure, contributing billions of dollars in value across the globe with more than 90% of commercial software containing one or more OSS project [1], [2]. Despite the growing importance and value of OSS, it continues to face significant challenges that threaten its long-term sustainability. For an OSS ecosystem to stand the test of time, projects must remain stable and robust in the face of changing contributors, processes, codebases, and evolving needs [3], [4]. Ensuring that knowledge and practices persist over time is essential to maintaining and sustaining project quality. Prior work has examined several dimensions of OSS governance, including how code contributions are committed to repositories, how pull requests and issues are managed, and how newcomers are on-boarded into a project [5]–[8].

However, while documentation serves as a essential medium through which coordination and governance norms are communicated within OSS ecosystems, no study has yet examined how health documentation [9], a crucial coordination and governance artifact in OSS projects, is itself governed and coordinated across projects. This is particularly concerning given that documentation such as README.MD and CONTRIBUTING.MD serve as critical entry points for newcomers [10], [11], while other artifacts such as SECURITY.MD, CODE_OF_CONDUCT.MD, and GOVERNANCE.MD [9] play a vital role in facilitating knowledge sharing around processes, governance, and policies ultimately reducing coordination friction across OSS ecosystems.

More specifically, this documentation has been studied largely from a static perspective focusing primarily on availability and quality dimensions such as whether it adequately addresses developers' needs, its correctness, currency, and usability [12]. This static lens creates a critical gap in our understanding of how documentation maintenance, coordination, and governance unfold as measurable, dynamic processes over time.

To fill this gap, we conducted an empirical study to investigate how OSS projects maintain crucial documentation [9] over time, examining maintenance rhythm, the intention behind maintenance, and the contributors who drive it. These three dimensions, established in the OSS governance literature [13], characterize governance in terms of coordination processes and contributor activity which are shaped by community norms. We operationalize these components as rhythm, ownership, and intention, respectively. Rhythm captures how documentation coordination processes recur over time, ownership reflects who holds authority over documentation, and intention distinguishes whether documentation changes are deliberate. Each dimension is also grounded in prior empirical work. Rhythm [4], [14], ownership [15], [16], and intention [17], we treat them as the measurable components of a single governance duality already established in theory, applied here to health documentation.

Our investigation was guided by the following research questions:

- **RQ1** What documentation rhythm patterns characterize OSS repositories, and how are these patterns distributed across projects?
- **RQ2** To what extent do OSS contributors exhibit intentional documentation behavior?
- **RQ3** How is documentation ownership distributed across contributors, and does this ownership structure predict downstream project outcomes?
- **RQ4** How do documentation rhythm, intention, and ownership evolve over a decade of project maturity?

Our findings surface four central results. We identify three distinct documentation rhythm archetypes, Consistent, Occasional, and Sparse, showing that when projects document at all, they typically do so in irregular bursts rather than as a sustained, ongoing practice **(RQ1)**. Using a novel commit-level intention taxonomy (DocOnly, DocDominant, DocNon-Dominant), we show documentation activity is rare but deliberate when it occurs **(RQ2)**. Applying bus factor analysis to health documentation, we find ownership structurally fragile, and show this structure is not only descriptive: documentation bus factor independently predicts documentation staleness, and documentation participation breadth independently predicts newcomer retention, both beyond project size and activity volume **(RQ3)**. Finally, tracking each dimension annually across the decade reveals that rhythm archetypes are durable and self-reinforcing once established, Consistent and Sparse Documenters persist as the same archetype year over year 76% and 62% of the time respectively, while the deliberateness and breadth of documentation engagement have each declined significantly over the same window **(RQ4)**.

II. Related Work

Documentation in software engineering and OSS: The importance of good software documentation cannot be overstated, it is fundamental to the use, maintenance, adoption, and evolution of any software system [18]. This holds equally true for OSS projects, where documentation quality has been extensively studied and tools and methods have been developed to assess it [12], [19]. Among the most studied artifacts is the `README.md` file, which typically serves as the primary entry point for users visiting a repository [11]. Given its visibility and influence, numerous methods and techniques have been proposed to improve its creation, quality, and maintenance [10], [20]. Beyond the `README.md`, other project health documentation artifacts play equally important roles across different dimensions of OSS project health. Files such as `Contributing.md`, `Security.md`, `Code _of _Conduct.md`, and `Governance.md` govern onboarding, security practices, community behavior, and project governance, respectively, and have attracted growing research interest, with methods and techniques proposed to improve and maintain them [21]–[23]. These prior contributions are largely motivated by evidence that the quality, accessibility, freshness, and readability of such documentation are key factors in successful contributor onboarding, security posture, governance effectiveness, and the overall sustainability of OSS ecosystems [7], [11], [24]–[26].

Project Health, Coordination and Governance in OSS: Project health metrics provide measurable insights into the overall state of an OSS project, serving as valuable signals for adopters, downstream consumers, and supply chain utilization decisions. Key indicators such as commit activity, issue resolution rates, pull request merge rates, bus factor and contributor churn and retention have been widely studied as proxies for project vitality and long-term sustainability [4], [27]. Unlike traditional software development, where coordination and governance are typically enforced through formal organizational structures, OSS ecosystems operate under fundamentally different dynamics. As OSS adoption has grown, understanding how coordination and governance function within these ecosystems has become increasingly critical particularly how the relationships between contributors influence project sustainability, a dimension that has been extensively studied through a socio-technical lens [5], [28]. Shaikh et al. [13] investigated the duality of coordination and governance in OSS, finding that both operate in parallel and are mutually reinforcing. Their work further characterizes governance as encompassing shared rules, norms, and instructions, alongside activities that operationalize the authoritative structure of an ecosystem with common coordination patterns observable directly from version control system activity. Beyond contributor interactions, documentation artifacts serve as a complementary mechanism for facilitating coordination and governance in OSS projects [9], codifying the norms and expectations that guide how contributors engage with and sustain the ecosystem over time. These efforts, however, characterize governance at a single point in time; whether the resulting patterns are stable properties of a project or drift as it matures over a multi-year horizon remains largely unexamined, a gap prior temporal work on OSS commit rhythms [4], [14] has begun to address for code but not documentation. We address this directly by tracking documentation rhythm, intention, and ownership annually across a full decade (RQ4).

III. Dataset Construction

Repository Selection: For this study, we selected the top 100 repositories based on the following criteria, which we deemed most relevant to our research questions:

Community Interest (Stars $>$ 10,000): We prioritized repositories with high star counts as a proxy for sustained community interest and visibility [29], reflecting significant engagement and adoption across the developer community over time.

Maturity ($\geq$ 10 Years): We selected repositories that are at least a decade old, as they provide sufficient historical depth to observe stable maintenance patterns, long-term evolutionary trends, and well-established governance practices.

Active Adoption (Forks $>$ 9,000): We focused on repositories with substantial fork counts, as forking activity reflects real-world usage and downstream adoption within the broader developer community [30].

Software Development Focus: We excluded repositories primarily serving educational purposes including those containing books, programming tutorials, videos, and instructional materials to ensure our dataset reflects genuine software development dynamics rather than curated learning content.

Active Status: We excluded archived repositories during manual filtering to ensure our dataset consists of currently active projects that continue to evolve, attract contributions, and undergo meaningful maintenance activity.

Activity Window (2016-2026) : We use the most recent decade as the observation window to estimate documentation activity patterns in a comparable mature phase across repositories.

Project Health Documentation: We define **project health documentation** as repository-residing documentation artifacts that directly support project governance, contributor coordination, and community continuity. In contrast to other technical documentation (e.g., API references, tutorials, or feature-specific manuals), health documentation captures the operational and social infrastructure of a project: how contributors enter, participate in, and align with project norms. These artifacts, labeled as community health documents by GitHub [9], support stable coordination processes that shape on-boarding, participation expectations, reporting pathways, and maintenance practices, making them especially relevant to the study of documentation governance and OSS sustainability. The specific documentation included in our analysis are shown in Table I.

TABLE I
HEALTH DOCUMENTATION ARTIFACTS

Artifact	Description
`README`	Primary entry point and project overview
`CONTRIBUTING`	Guidelines for contributing
`CONTRIBUTORS`	List of project contributors
`COMMIT_CONVENTIONS`	Commit message standards
`BUILDING`	Build instructions
`PULL_REQUEST_TEMPLATE`	Template for pull requests
`ISSUE_TEMPLATE`	Template for issue reporting
`CHANGELOG`	Log of changes across versions
`HISTORY`	Historical record of changes
`RELEASE(S)`	Release notes and announcements
`CODE_OF_CONDUCT`	Community behavior standards
`GOVERNANCE`	Governance structure and policies
`SUPPORT`	Guidance for obtaining support
`MAINTAINERS`	List of active maintainers
`SECURITY`	Vulnerability reporting policy
`LICENSE`	Project license terms
`NOTICE`	Legal notices and attributions
`COPYING`	Copying and distribution terms
`AUTHORS`	Official list of project authors
`CREDITS`	Contributor acknowledgments
`THANKS`	Recognition of supporters
`ROADMAP`	Planned development directions
`VISION`	Long-term goals and philosophy

Data Extraction: We locally cloned all 100 repositories selected based on our inclusion criteria (Section III) and extracted documentation-related commits from their version histories using native Git commands. For each repository, we collected commit-level metadata, including the commit hash, author identity, timestamp, commit message, the specific health documentation files modified in each commit, and each commit's automated-vs-human authorship status to support bot filtering and identity normalization (Section IV-C). These data enabled our analysis of documentation maintenance rhythm, intention, and ownership. Automated actors are excluded from all ownership and structural-fragility calculations throughout; where rhythm metrics are affected by automated commit activity, we report a dedicated robustness check (Section IV-A, Artifact-Type and Bot Robustness).

IV. METHODOLOGY

A. *Documentation Rhythm Patterns, Characteristics, & Distribution (RQ1)*

Motivation and Problem Addressed: Existing OSS health metrics [31] treat documentation as a binary or volumetric construct (e.g., either a project has documentation or it does not) where more is generally assumed to be better. Particularly, measures of documentation tend to focus on file count, coverage measurement, or quality assessment at a single point in time. What none of these approaches capture is the temporal dimension, which is whether documentation activity is sustained, regular, and deliberate over time, or whether it happens in isolated bursts separated by long periods of neglect. Documentation maintenance rhythm refers to the temporal regularity with which a project sustains documentation activity over time. We further evaluate documentation maintenance rhythm at a monthly granularity, which prior work identified as a feasible and meaningful unit of analysis for understanding maintenance patterns within OSS ecosystems [14]. Building on this, we operationalize documentation maintenance rhythm across 121 months spanning a decade-long observation period using normalized shannon entropy. Additionally, we evaluate an active window rate, defined as the proportion of months within the observation period in which at least one documentation maintenance commit was recorded, capturing the breadth of sustained documentation activity across the evaluation window.

Normalized Shannon Entropy: Let $x_1, x_2, \ldots, x_M$ denote the number of health documentation related commits observed in each monthly window over the analysis period, where $M = 121$ is the total number of months in the observation window.

The Shannon entropy is computed as

$$H = -\sum_{i=1}^{M} p_i \log_2 p_i,$$

where terms with $p_i = 0$ are treated as contributing 0 (consistent with the standard convention $\lim_{p\to 0} p \log_2 p = 0$). We normalize entropy using it's theoretical maximum to account for the length of the observation window.

$$H_{\text{norm}} = \frac{H}{\log_2 M}.$$

This yields $H_{\text{norm}} \in [0, 1]$, where values approaching 1 indicate documentation activity distributed evenly across months, and values approaching 0 indicate activity concentrated in few months.

Active Window Rate (AWR): Let x_i denote the number of health-file documentation commits in month i, for $i = 1, \ldots, M$, using the same observation window as defined above ($M = 121$ months). We define an indicator function for whether a month contains any documentation activity:

$$\mathbf{1}(x_i > 0) = \begin{cases} 1, & \text{if month } i \text{ contains at least one} \\ & \text{health-file documentation commit,} \\ 0, & \text{otherwise.} \end{cases}$$

The Active Window Rate is then defined as

$$AWR = \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}(x_i > 0).$$

$AWR \in [0, 1]$, where $AWR = 1$ indicates documentation activity in every observed month and $AWR \approx 0$ indicates near-complete absence of documentation activity across the window. AWR measures the continuity of documentation presence across the full window capturing whether extended documentation gaps exist regardless of how active the project is when it does document.

Maintenance Patterns across OSS Ecosystem: To reveal maintenance rhythms across the OSS ecosystems we utilize K-means Clustering [32] with $n\_init = 10$ random initializations which retains the solution within the lowest inertia to mitigate sensitivity to random seed selection. A fixed random seed (42) was also used to ensure reproducibility.

Determining the Number of Clusters: To determine the number of clusters k we used the silhouette score [33], which measures the cohesion and separation of cluster assignments on a scale of $[0, 1]$, where higher values indicate more well-defined clusters. We report silhouette scores for the combinations involving entropy and active window rate. Following the K-means clustering fitting, we assigned archetypes interpretable names that reflect their rhythm quality.

B. Documentation Behavior Intent (RQ2)

Motivation and Problem Addressed: When documentation maintenance occurs, it is not always clear whether these efforts were deliberate. Consider two commits that both modify README.MD: in the first, a contributor opens a dedicated commit to update the README after a user files an issue reporting outdated installation instructions. This represents intentional, standalone documentation work. In the second, the README is touched incidentally alongside dozens of source files as part of a broader refactor, with the documentation change bundled into code work. Both appear identical in a commit log, yet reflect fundamentally different signals about how a project values documentation as a practice. No existing metric distinguishes intentional documentation behavior from incidental documentation behavior at the commit level. We propose a ***documentation intention*** metric that addresses this gap by examining the ratio of documentation-primary commits to all documentation-touching commits and capturing whether documentation effort is dedicated or incidental.

Documentation Intention Formulation: Let $\mathcal{H}$ denote the set of health documentation file paths as defined in Section III (README, CONTRIBUTING, CHANGELOG, SECURITY, and related governance files). We define T as the set of commits in the observation window (2016-2026) that modify at least one file in $\mathcal{H}$:

$$T = \left\{ c \mid \exists f \in \mathcal{H} : f \in \text{files}(c) \right\}$$

All intention metrics are computed over T. Commits that modify only non-health documentation files carry no documentation intent signal and are excluded from this analysis. Let $N_T = |T|$, For each commit $c \in T$, we define:

i. $h(c)$: number of health documentation files modified in c
ii. $o(c)$: number of non-health documentation files modified in c.

The documentation share of commit c is:

$$s(c) = \frac{h(c)}{h(c) + o(c)}$$

Using a proposed dominance threshold τ ($\tau = 0.5$), The threshold $\tau = 0.5$ reflects the natural majority boundary commits where health documentation files exceed half of all modified files are, by definition, documentation-primary. Each commit $c \in T$ is assigned to exactly one of three mutually exclusive categories:

$$c \in \textit{DocOnly} \iff o(c) = 0$$

$$c \in \textit{DocDominant} \iff o(c) > 0 \;\wedge\; s(c) \geq \tau$$

$$c \in \textit{DocNonDominant} \iff o(c) > 0 \;\wedge\; s(c) < \tau$$

These three categories form a complete partition of T:
$$T = \textit{DocOnly} \cup \textit{DocDominant} \cup \textit{DocNonDominant}$$

DocOnly commits modify exclusively health documentation files, representing the strongest signal of deliberate documentation intent. **DocDominant** commits touch other files but remain documentation focused, with health documentation files comprising at least half of all modified files. **DocNonDominant** commits modify documentation files incidentally, documentation is a minor component of a broader change.

C. Documentation Ownership, Contributor Concentration, Sustainability, & Outcomes (RQ3)

Motivation and Problem Addressed: The bus factor literature focuses on code contributions. Studies [15], [16] have developed methods to measure contributor concentration in code, but documentation is systematically excluded from these analyses. Documentation commit volume and intention, as characterized in RQ1 and RQ2, reveal when and how documentation is maintained, but not *by whom*. RQ3 examines the ownership structure of documentation activity, that is, the breadth of contributor participation, the degree of concentration among active documentation contributors, and the structural fragility that concentration implies.

Contributor Identification: Two contributor pools are defined for each repository r:

i. $C_{all}(r)$: the set of all unique human contributors who made at least one commit to r within the observation window, regardless of file type.
ii. $C_{doc}(r)$: the set of contributors who contributed to at least one health documentation file T defined in Section IV-B

C_{all} is extracted from all commits using git log from the cloned repositories and C_{doc} from the health documentation-touching commit set T defined in Section IV-B. We used several bot filtering patterns (i.e., 34) ensuring that automated the exclusion of bots from both $C_{all}(r)$ and $C_{doc}(r)$ pools consistently. Identity normalization was also applied to resolve cases where the same contributor appears under multiple identities. GitHub noreply addresses of the form N+username@users.noreply.github.com were normalized to username@users.noreply.github.com, merging commits made through the GitHub web editor with those made via a local git client by the same contributor.

Documentation Contributor Participation A contributor is counted in C_{doc} if they touched at least one health documentation file at least once, regardless of how many documentation commits they authored. It therefore captures the *breadth* of participation rather than depth, establishing what fraction of the contributor pool ever engages with health documentation at all. We report documentation contributor participation at both the per-repository level and as an ecosystem aggregate.

Ownership Concentration: To measure how evenly documentation commit activity is distributed among contributors C_{doc}, that is, the percentage of documentation commits the top contributors in a repository have made. We compute the top-k commit share for $k \in \{1, 3, 5, 10\}$. Reporting four values of k reveals how rapidly concentration accumulates: a project where $TopK(r, 1) \approx TopK(r, 5)$ is effectively a single-contributor documentation project regardless of how many contributors nominally appear in C_{doc}.To establish whether documentation commits are more concentrated than general commit activity, we also compute the same top-k share over C_{all} using all commits (code, configuration, and documentation combined).

Structural Fragility: Top-k share measures concentration as a continuous property. To operationalize the *risk* implied by that concentration, we adopt the bus factor metric [15], defined as the minimum number of contributors whose combined activity accounts for at least a given fraction x of total documentation commits. We report $Bus_{0.5}$ (Bus-50) and $Bus_{0.8}$ (Bus-80). Bus-50 captures the critical fragility threshold, the point at which the departure of those contributors would remove majority control over health documentation governance. Bus-80 captures the broader fragility envelope, identifying how many contributors the project would need to lose before documentation activity effectively collapses. Both metrics are computed over C_{doc}.

Documentation and Repository-Wide Outcomes: Structural characterization alone does not establish that participation breadth or ownership concentration matter beyond description. We therefore test both constructs against the following outcomes:

i. **Documentation staleness**: the number of days between a repository's last health-documentation commit and the end of the observation window, aggregated per repository as the median across its health documentation files; higher values indicate documentation that has gone longer without being revisited.
ii. **Documentation-contributor retention**: the proportion of a repository's health-documentation newcomers, contributors whose first documentation touch falls within the observation window, who touch documentation again in a second, distinct month.
iii. **Overall activity trend**: the slope of a linear fit to monthly commit volume across the full decade, testing whether documentation governance predictors say anything about a repository's trajectory beyond documentation itself. We test several analogous repository-wide outcomes (general contributor retention, newcomer arrival, contributor growth, maintainer turnover) using the same procedure.

For each governance predictor (documentation participation rate, documentation bus factor, entropy, and active window rate) and each outcome, we fit a nested regression, outcome on log-transformed contributor count alone, versus contributor count plus the predictor. We additionally require the result to survive three further checks before we treat it as established:

i. the predictor must retain significance when raw documentation-commit volume is added as a second control, distinguishing breadth-of-participation effects from simple effort-volume effects;
ii. the result must hold under heteroskedasticity-robust (HC3) standard errors; and
iii. the result must hold after removing the three most influential repositories by Cook's distance.

We report a predictor outcome relationship as validated only when it survives all four checks size, volume, robust, and outlier.

D. Documentation Governance Evolution (RQ4)

Motivation and Problem Addressed: RQ1–RQ3 characterize documentation rhythm, intention, and ownership as they stand across our decade-long observation window, but say nothing about whether these dimensions are stable properties of a project or shift as it matures. A rhythm archetype, intention rate, or ownership structure measured once could reflect a durable trait or a transient snapshot; only repeated, decade-long observation can distinguish the two, and no prior study of OSS documentation has had the observation window to do so. RQ4 asks whether each dimension is stable or shifting over time, and if shifting, in which direction.

Annual Archetype Classification: To test rhythm stability, we classify each repository's rhythm independently in each of the 10 years spanning the observation window, computing entropy and active window rate within each 12-month slice and assigning the year to the nearest of the three archetype centroids established in RQ1 (Section IV-A). We hold the archetype boundaries fixed at their decade-derived values rather than re-clustering within each year, so that an observed shift reflects a genuine change in a repository's rhythm rather than drift in the reference boundaries themselves. Repository-years with fewer than 3 health-documentation commits (14.7%

of repository-years) are excluded from classification as too sparse to support a reliable rhythm estimate. From the resulting per-repository-year labels we construct two transition analyses: an *adjacent-year* transition matrix pooling all consecutive valid year-pairs across all repositories, and a coarser *decade-span* transition comparing each repository's earliest to latest valid year (restricted to repositories with at least two valid years). Both are row-normalized to report, for each starting archetype, the proportion of transitions landing in each destination archetype.

Annual Trend Testing: To test whether intention (Section IV-B) and ownership breadth (Section IV-C) shift over time, we compute the documentation-only rate and documentation participation rate separately for each repository-year, then apply two complementary trend tests: a Spearman rank correlation between year and metric value across all raw repository-year observations, and a Mann-Kendall trend test (Hamed-Rao modified variant, robust to serial correlation) on the annual cross-repository median. The two tests differ in what they are sensitive to, Spearman weights every repository-year observation individually, while Mann-Kendall tests the trend in central tendency across years, so agreement between them indicates a trend that is not an artifact of repository-level heterogeneity or of a specific averaging choice.

V. RESULTS

A. *Documentation Rhythm Pattern (RQ1)*

Metric Distribution: The median entropy (0.805) is higher than the mean (0.787), indicating a left-skewed distribution. Most repositories cluster toward higher entropy values but a tail of low-entropy repositories pulls the mean downward. We also observed that the active window rate has a wide spread ($SD = 0.254, range = [0.058, 1.000]$), which suggests substantial variation in how persistently repositories maintain documentation presence across a full decade. Table II summarizes the descriptive statistics of the two rhythm metrics (Shannon Entropy and Active Window Rate) computed across all 100 repositories.

TABLE II
DESCRIPTIVE STATISTICS OF RHYTHM METRICS

Metric	Mean	Median	SD	Min	Max
Entropy (H_{norm})	0.787	0.805	0.126	0.406	0.989
Active Window Rate	0.547	0.508	0.254	0.058	1.000

Documentation Rhythm Archetypes: The K-Means clustering shown in Figure 1 shows the silhouette scores for the k-test values with ($k = 3$, silhouette $= 0.563$) identified as the optimal number of archetypes across the 100 repositories, computed over the full decade of available history for each repository. Following cluster fitting, we assigned repository archetype names based on entropy rank, with both entropy metrics and active window rate producing identical cluster orderings. The resulting archetypes were ***Consistent Documenters*** ($n = 37$, 37%), ***Occasional Documenters*** ($n = 50$, 50%), and ***Sparse Documenters*** ($n = 13$, 13%). Table III shows the summarized distribution of the archetypes across metrics.

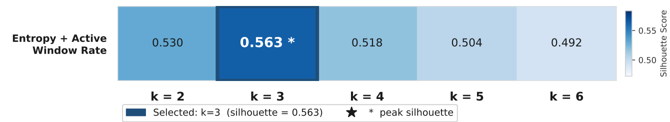

Fig. 1. Silhouette Selection of K

Consistent Documenters ($n = 37$, 37%) maintain high documentation rhythm across both dimensions with evenly distributed activity (mean $H_{\text{norm}} = 0.906$) and persistent presence across the observation window (mean $AWR = 0.825$). Documentation was present in approximately 83% of observed months on average. **Occasional Documenters** ($n = 50$, 50%) represent the largest and most prevalent archetype, characterized by moderate entropy (mean $H_{\text{norm}} = 0.762$) and notably lower active window rate (mean $AWR = 0.440$), indicating documentation activity concentrated in approximately 44% of observed months. Activity occurs in periodic bursts rather than as a continuous practice, with extended gaps between documentation efforts. **Sparse Documenters** ($n = 13$, 13%) exhibit near-absent documentation rhythm, with low entropy (mean $H_{\text{norm}} = 0.546$) and minimal active window rate (mean $AWR = 0.167$), indicating documentation activity in approximately one month in six on average.

TABLE III
DESCRIPTIVE STATISTICS FOR EACH DOCUMENTATION RHYTHM ARCHETYPE ACROSS METRICS.

Archetype	n	Metric	Mean	Median	Min	Max
Consistent	37	H_{norm}	0.906	0.908	0.832	0.989
		AWR	0.825	0.818	0.620	1.000
Occasional	50	H_{norm}	0.762	0.758	0.647	0.848
		AWR	0.440	0.450	0.256	0.628
Sparse	13	H_{norm}	0.546	0.534	0.406	0.637
		AWR	0.167	0.174	0.058	0.355

The artifacts in Table I span very different maintenance cadences, from actively-evolving process documentation to write-once legal and attribution files, so we validated the archetype construct against this heterogeneity before treating it as the primary RQ1 result. We partition the 22 artifact types into **living/process** and **static/attribution** documentation (full mapping in Table IV) and recompute entropy and AWR within each stratum. At the ecosystem level, 92.2% of documentation-touching commits touch only living/process files, and combined-type entropy and AWR correlate with their living-only counterparts at $\rho = 0.93$ and $\rho = 0.94$ (Spearman), respectively, confirming the reported archetypes are driven overwhelmingly by living/process documentation rather than by the inclusion of rarely-touched static files.

TABLE IV
ARTIFACT-TO-CATEGORY MAPPING USED IN THE LIVING/STATIC ROBUSTNESS CHECK (RHYTHM TYPE) AND THE PURPOSE TAXONOMY (PURPOSE).

Dimension	Category	Artifacts
Rhythm type	Living/process	README, CONTRIBUTING, CHANGELOG, HISTORY, RELEASE(S), GOVERNANCE, SECURITY, SUPPORT, MAINTAINERS, CODE_OF_CONDUCT, ROADMAP, VISION, BUILDING, COMMIT_CONVENTIONS, PULL_REQUEST_TEMPLATE, ISSUE_TEMPLATE
	Static/attribution	LICENSE, NOTICE, COPYING, AUTHORS, CREDITS, THANKS, CONTRIBUTORS
Purpose	Onboarding/process	README, CONTRIBUTING, BUILDING, COMMIT_CONVENTIONS, PULL_REQUEST_TEMPLATE, ISSUE_TEMPLATE
	Governance/policy	GOVERNANCE, CODE_OF_CONDUCT, SECURITY, SUPPORT, MAINTAINERS, ROADMAP, VISION
	Change-tracking	CHANGELOG, HISTORY, RELEASE(S)
	Legal/attribution	LICENSE, NOTICE, COPYING, AUTHORS, CREDITS, THANKS, CONTRIBUTORS

Documentation Purpose Taxonomy: To characterize what "documentation maintenance" concretely consists of, we further partition the 22 artifact types into four purpose-built categories, *onboarding/process*, *governance/policy*, *change-tracking*, and *legal/attribution* (mapping in Table IV). Change-tracking accounts for the largest ecosystem-level share of commits, but this is driven by a small number of repositories with heavy, often automated, changelog activity: the median repository has *zero* change-tracking commits across the full decade-long window. Onboarding/process documentation, by contrast, is the typical governance activity: it is present in every repository in the sample and accounts for a median 84.6% of a repository's documentation commits. Governance/policy and legal/attribution activity are minor and frequently absent altogether. Table V shows that the ecosystem-level and per-repository pictures diverge sharply, and the divergence is itself informative.

TABLE V
DOCUMENTATION PURPOSE TAXONOMY: ECOSYSTEM-LEVEL SHARE, MEDIAN PER-REPOSITORY SHARE, AND REPOSITORIES WITH ZERO ACTIVITY, BY CATEGORY.

Category	Ecosystem	Median	Zero activity
Change-tracking	50.8%	0.0%	54/100
Onboarding/process	41.3%	84.6%	0/100
Legal/attribution	7.7%	3.9%	11/100
Governance/policy	3.4%	3.3%	25/100

Summary — RQ1

Three documentation rhythm archetypes emerged across 100 OSS repositories: *Consistent Documenters* (37%), *Occasional Documenters* (50%), and *Sparse Documenters* (13%). The majority of repositories exhibit occasional rather than consistent documentation rhythm, with documentation activity present in only 44% of observed months on average for the largest archetype.

B. Documentation Intention (RQ2)

Documentation Intention: Across the 100 repositories, health documentation commits constitute a median of only 1.90% of total commit activity (mean = 5.78%). Consequently, we examine how intentionally this documentation activity occurs.

Doc-Only Commits: Across the 37,447 documentation-touching commits in the decade-long dataset, 29,171 (77.9%) were documentation only, making this the dominant commit category at the ecosystem level. At the per-repository level, the documentation-only rate exhibited a left-skewed distribution (**mean** $= 0.785$**, median** $= 0.825$**, SD** $= 0.172$), indicating that most repositories are concentrated toward the higher end of the scale.

Doc-Dominant and Doc-Non-Dominant Commits: Documentation dominant mixed commits modify both health documentation files and other files, but with documentation files constituting at least half of all modified files ($\tau = 0.5$). At the ecosystem level, 4,564 commits (12.2% of T) were classified as documentation-dominant. Documentation non-dominant commits, which modify health documentation files as a minor component of a broader change, accounted for the remaining 3,712 commits (9.9% of T), the smallest of the three categories. Taken together, documentation-only commits dominate at 77.9% of all 37,447 documentation touching commits over the decade, followed by documentation-dominant mixed commits at 12.2% and non-dominant incidental commits at 9.9%. Considered alongside the baseline finding that health documentation constitutes a median of **only 1.90% of total commit activity**, these proportions establish the central finding that health documentation commits are rare in the context of overall development effort, but when they occur, they are deliberate. This deliberateness is not static, however; we find it has eroded significantly over the decade studied (Section V-D).

Summary — RQ2

When documentation is touched, 90.1% of those commits are documentation-only or documentation-dominant. Deliberate documentation behavior consistently outweighs incidental contributions, though the documentation-only share has significantly declined over the decade studied. The core governance challenge is not a lack of intent; contributors document purposefully when they engage. It is rather a lack of frequency, and, increasingly, a lack of intentional documentation practice as projects mature.

C. Documentation Ownership (RQ3)

Documentation is a Structural Minority Practice: Across the 100 repositories, only 7,988 of 179,622 total contributors (4.5% at the ecosystem level) ever touched a health documentation file within the decade-long observation window. At the per-repository level, the median participation rate is **6.3%** (mean $= 9.9\%$, SD $= 0.119$, IQR $= 2.9\%$–10.9%), which implies that the typical project relies on a small minority of its contributors for health documentation governance. Forty-four repositories recorded participation rates below 5%, and only four exceeded 50%. The raw counts reflect this disparity directly: repositories averaged 1,796 total contributors (median $= 1,025$) but only 79.9 documentation contributors (median $= 46$). This participation rate has itself declined significantly over the decade studied (Section V-D).

Documentation Ownership is More Concentrated Than Code Ownership: Documentation commits are more concentrated than general commit activity across every top-k threshold examined. The mean top-3 share for documentation commits is **0.517** compared to **0.415** for all commits, with 79 of 100 repositories showing higher documentation concentration. The gap persists at higher k: top-5 (0.600 vs. 0.484) across 80 repositories and top-10 (0.705 vs. 0.577) across 82 repositories. For top-1 share, the single largest documentation contributor accounts for a median of **25.2%** of all documentation commits in their repository (mean $= 32.2\%$). reflecting a subset of projects with extreme single-contributor dependence even across a full decade of activity.

Documentation Governance is Structurally Fragile: The documentation bus factor (Bus-50) has a median of **4** across the 100 repositories (mean $= 8.54$, SD $= 19.53$, IQR $= 2$–7). Forty-seven repositories (47%) have a Bus-50 of three or fewer, this implies that the departure of at most three contributors would shift majority documentation control. More critically, **15 repositories (15%) depend on a single contributor** for the majority of their documentation activity (Bus-50 $= 1$) even across a full decade. Bus-80 confirms the broader fragility, the median repository requires only 16 contributors to account for 80% of documentation commits, and 35 of repositories (35%) reach that threshold with 10 or fewer. Bus-50 correlates strongly with total contributor count ($\rho = +0.58$, $p < 0.0001$), confirming that larger repositories naturally attract more documentation contributors and distribute ownership more broadly, and neither entropy nor active window rate add explanatory power over contributor count alone, consistent with our RQ1 finding that documentation rhythm and ownership concentration are largely independent dimensions of governance.

Ownership Structure Predicts Documentation Outcomes Documentation rhythm archetype shows strong bivariate associations with general contributor retention, newcomer arrival, contributor growth, maintainer turnover, and overall activity trend outcomes, but every one is fully explained by contributor ecosystem size and adds no explanatory power in a confound-controlled test. Three relationships, however, survive the full four-check (Table VI).

TABLE VI
CONFOUND-CONTROL (p-VALUES) FOR THE THREE VALIDATED PREDICTOR–OUTCOME RELATIONSHIPS.

Predictor → Outcome	Size	Volume	Robust	Outlier
Bus factor → Staleness	0.0015	0.0015	0.0038	0.0031
Participation → Retention	0.0078	0.0377	0.0003	0.0053
	0.0034	0.0220	0.0002	0.0018
Participation → Activity trend	0.027	0.009	0.019	0.043

Bus factor predicts documentation freshness. Documentation bus factor predicts staleness across all four checks in Table VI: repositories with a *higher* bus factor, documentation responsibility spread across more contributors rather than concentrated in one or two, keep their documentation measurably fresher (partial $\rho = -0.394$, a moderate negative association; significance comes from the four checks, not from the size of ρ alone). Entropy and active window rate show a similar bivariate pattern but do not clear the battery once documentation volume is controlled, and participation rate alone does not survive robust standard errors ($p = 0.090$). Ownership concentration, not the raw volume or rhythm of documentation activity, is what most reliably distinguishes repositories that keep documentation current from those that do not.

Participation breadth predicts newcomer retention. A construct-matched test, predicting whether documentation newcomers return to touch documentation again rather than a domain-mismatched general-activity outcome, shows documentation participation rate predicts newcomer retention across all four checks, strengthening further when restricted to repositories with at least 10 documentation newcomers. Bus factor's relationship with retention does not clear the same battery in either the contemporaneous or a temporally-lagged design (bus factor computed on only the first half of the observation window, predicting established-contributor loss in the second half), which rules out reverse causality as the explanation.

Participation breadth predicts overall project activity, beyond documentation itself. Documentation participation rate also independently predicts the repository's general activity trend, the slope of overall monthly commit volume across the decade, not just documentation-specific outcomes. Repositories where documentation involvement is spread across more contributors show a less-declining, or more-growing, trajectory of overall project activity, beyond what project size and documentation effort alone predict.

Scope of the validated claims: Ownership structure carries genuine predictive value, but its two facets specialize: concentration (bus factor) predicts documentation currency, while breadth (participation rate) predicts both who stays engaged with documentation and the trajectory of the project's overall activity.

Summary — RQ3

Health documentation is maintained by a minority of contributors, and that minority has been shrinking. The median repository has only 6.3% of its contributor pool engaging with health documentation (ecosystem rate $= 4.5\%$), a rate that has declined significantly over the decade studied. Ownership structure carries genuine predictive value: documentation bus factor independently predicts documentation staleness, documentation participation breadth independently predicts newcomer retention, and also participation breadth independently predicts a repository's overall activity trend, all beyond project size and activity volume.

D. Documentation Governance Evolution (RQ4)

RQ1–RQ3 characterize documentation governance as it stands across the decade studied. Using the annual classification and trend-testing procedures described in Section IV-D, we test whether each dimension, rhythm, intention, and ownership, is stable or shifting over time.

Rhythm is durable: Table VII reports the pooled adjacent-year transition matrix across all 710 valid consecutive year-pairs. Consistent Documenters remain Consistent the following year 76.3% of the time and Sparse Documenters remain Sparse 61.7% of the time; Occasional Documenters persist only 37.6% of the time, more often drifting toward Sparse (43.1%) than remaining Occasional. Comparing each repository's earliest to latest valid year ($n = 98$ repositories with at least two valid years) shows the same ordering at a coarser grain: 46.8% decade-long persistence for Consistent, 73.3% for Sparse, and only 33.3% for Occasional. Consistent and Sparse behave as self-reinforcing, sticky states, while Occasional is a transitional state repositories pass through rather than settle into, and one that drains toward Sparse far more often than it climbs toward Consistent.

TABLE VII
ADJACENT-YEAR RHYTHM ARCHETYPE TRANSITION MATRIX (ROW-NORMALIZED %, POOLED ACROSS 710 VALID CONSECUTIVE YEAR-PAIRS). ROWS ARE THE ARCHETYPE IN YEAR y, COLUMNS THE ARCHETYPE IN YEAR $y+1$.

From ↓ To →	Consistent	Occasional	Sparse	n
Consistent	76.3%	18.7%	5.0%	299
Occasional	19.3%	37.6%	43.1%	202
Sparse	6.7%	31.6%	61.7%	209

Intentionality is eroding: The documentation-only rate declines significantly year over year (Spearman $\rho = -0.076$, $p = 0.027$; Mann-Kendall $p = 0.003$), with the annual median falling from 87.7% in the first year of the window to 75.0% in the final year, a 12.7-point decline.

Ownership breadth is narrowing: Documentation participation rate shows the same declining pattern (Spearman $\rho = -0.126$, $p = 0.0001$; Mann-Kendall $p = 0.0007$), with the annual median falling from 7.1% to 4.3% across the same window, a 2.8-point decline.

Rhythm archetypes entrench once established, while the breadth and deliberateness of documentation engagement both erode as projects mature; these are divergent trajectories along the same decade, the coarse label a project carries becomes more entrenched even as the people and deliberateness sustaining it thin out. Section VI discusses the implications of this combination alongside our predictive-validity results.

Summary — RQ4

Documentation governance is not static. Rhythm archetypes are self-reinforcing: once a repository becomes a Consistent or Sparse Documenter, it is more likely to remain one the following year (76% and 62% respectively), while Occasional functions as a transitional state that drains toward Sparse more often than it climbs toward Consistent. At the same time, both the deliberateness (DocOnly rate, -12.7 points) and breadth (participation rate, -2.8 points) of documentation engagement have eroded significantly over the same decade. Documentation governance is entrenching in its coarse rhythm while narrowing in who sustains it and how deliberately.

VI. DISCUSSION

Documentation Governance and Its Downstream Consequences: Documentation rhythm captures the currency of health artifacts over time; a project transitioning toward Sparse rhythm has governance documentation that increasingly fails to reflect current project norms, raising the onboarding barriers that prior work has linked to newcomer dropout and contributor attrition [7], [25]. Our outcome-validation results (Section V-C) substantiate this connection directly for ownership concentration rather than leaving it implicit: repositories with a lower documentation bus factor keep their documentation measurably fresher, controlling for both community size and how much documentation work is produced overall. Documentation bus factor connects to sustainability risk constructs associated with project abandonment [15], [16]; our finding that 15% of repositories have Bus-50= 1 for documentation, even measured across a full decade, indicates that governance fragility is widespread even among the most mature projects studied. Participation breadth, in turn, predicts a different outcome, newcomer retention, and this relationship survives a temporally-ordered check that concentration alone does not, suggesting these two facets of ownership structure are complementary rather than substitutable governance signals: concentration keeps documentation current, while breadth keeps new contributors engaged with it. Participation breadth's reach extends further still: it is also the one governance signal in this study that predicts a repository's overall activity trend, not just documentation-specific outcomes, indicating that who participates in documentation is informative about project health more broadly, not only about documentation itself. The decline in both documentation participation and documentation-only commit rates over the decade studied points to a broader structural shift: as OSS projects mature, documentation work becomes progressively more concentrated among fewer contributors and less often handled as a deliberate, standalone practice. Our evolution results (RQ4) sharpen the practical takeaway of this combination: because Occasional is a transitional rather than a stable state, and drains toward Sparse in the plurality of observed transitions, a repository's rhythm *trajectory*, not just its current archetype, is itself an early-warning signal. Tooling that flags a Consistent-to-Occasional shift could surface documentation risk before a project settles into the Sparse state our predictive-validity results (Section V-C) associate with elevated staleness, well before ownership concentration or participation loss would otherwise reveal the same risk.

Automated Documentation Synchronization

VII. Threats to Validity

A. Internal Threats

External Documentation: Our analysis was confined to repository resident observable health documentation related files. Manual inspection of the 100 repositories identified 20 projects that specifically host contributing related artifacts, a crucial health documentation on external websites or wikis rather than as in-repository files. This limitation is inherent in repository mining studies and in our study affects rhythm, intention, and ownership metrics uniformly across impacted repositories.

B. External Threats

Repository Selection and Generalization Our selection criteria requiring more than 10,000 stars, 9,000 forks, and ten years of history deliberately sample mature, high-visibility projects that benefit from large contributor communities and established governance cultures. Findings should not be generalized directly to smaller or younger repositories, where documentation fragility is likely more acute. We treat this as a conservative bound, if governance is structurally fragile even in the most successful OSS projects, the problem is likely worse across the ecosystem at large.

VIII. Conclusion and Future Work

This paper establishes documentation governance as a measurable, multi-dimensional, and evolving construct in OSS that is rare, deliberate, and structurally fragile, and shows that documentation bus factor independently predicts documentation staleness while documentation participation breadth independently predicts newcomer retention, both beyond project size and activity volume. Our decade-long observation window further reveals that documentation governance is not static: rhythm archetypes are durable and self-reinforcing rather than incidental, even as documentation participation and deliberateness have each declined significantly over the ten years studied. Several questions remain open. The relationships we identify are associational, not causal; establishing their causal structure, and specifically whether documentation ownership structure can be deliberately cultivated to improve both currency and retention simultaneously, is a natural next step, potentially through matched-repository or interventional designs. Extending the analysis to smaller and younger repositories will improve generalization.

A. Data Availability Statement:

Our replication package is made available: [34].

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